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ORIGINAL PAPER

# Products pricing and return strategies for the dual channel retailers

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## Abstract

This paper analyzed how different return strategies and return rates affect dual-channel retailers' profits and channel pricings. Return can stimulate sales; however, the return has presented significant challenges to retailers. The return has long been studied to maximize profit and pricing; however, the different return strategies for dual-channel retailers affect channel both. This paper aimed to study whether or not dual-channel retailers should allow customers to return items in two channels and whether or not the retailer should contract with the manufacturers and pay extra fees to return products. This study indicated when the retailer should allow customers' returns to maximize the profit by increasing the demand. It was discovered that when a customer's sensitivity factor for pricing is large (i.e., the demand is small) and the return rate is low, both retailer and manufacturer should object to contracting for handling returned products. However, when both the customer's sensitivity factor for pricing and the return rate are high, the retailer and the manufacturer should sign a contract to achieve maximum profit. Otherwise, the contract desire was only one-sided. The profit-maximizing retailers must balance the trade-off between the product demands, the return losses, and the return rates. This analytical work was verified with numerical simulation, and the results demonstrated implications for dual-channel retailers, return strategies, and pricing.

**Keywords** Dual-channel · Retailer and manufacturer · Return strategy · Pricing · Optimization

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## 1 Introduction

With the rapid development of the Internet and consumers' demand for diversity, dual-channel retailing becomes more appreciated by retailers. Macy's, Walmart, Best Buy, and Nordstrom adopted dual-channel retailing. With the vigorous development of dual-channel retail, the challenges become prominent. In 2019, the total amount of returned products in the U.S. retail market was $309 billion (https://nrf.com/). In 2020, the amount of returned products reached 428 billion, accounting for 10.6% of the total retail sales (https://apprissretail.com/), thus the loss caused by the returns to the retailers is tremendous, and returns have an extreme impact on the retailers' product pricing and profit.

Considering existing research, the cross-channel return is a relatively common strategy. Customers who buy offline can return products to any physical stores, and online customers can return products by mail back through online channels, or they may return products to any offline physical store (Dijkstra et al. 2019; Jin et al. 2020; Radhi and Zhang 2019). In practice, most retailers adopt cross-channel returns. For example, Best Buy promises customers can return any products purchased online in any of their physical stores (https://www.bestbuy.com/). Walmart also guarantees that if customers are not satisfied with the purchased product, they can return it by mail or return it to the physical store (https://www.walmart.com). However, none of the existing studies considered the impact of different return strategies on dual-channel retailers' pricing and profit. This study investigated the impact of return rates, using different channels, on the pricing,

Online retailers have an advantage, compared with offline retailers, when pricing products. However, when customers buy from online channels, they cannot see and touch products. As a result, the color, material, and size of the products may not satisfy customers' expectations. Then, the products will be returned. Customers are unable to touch the products through an online channel, thus some customers prefer to go to physical stores to purchase products and avoid returns because of the uncertainty (Dzyabura and Jgabathula 2018: Kim et al. 2019). When a consumer purchases a product in a physical store, he/ she can touch, feel, and experience various aspects of the product. Retailers also provide specific services—such as product explanations and recommendations—in the store. The probability of purchasing unsuitable goods is reduced when shopping in-person, and that reduces the probability of returning goods. Conversely, offline channels need to bear higher marginal costs than online retailers. With increased return rates, it is costly for retailers to process returned products. Making a contract with the manufacturer (hereafter retailer was referred to as *she* and manufacturers were referred to as *he*) to handle returned products is a potential method to avoid return losses.

To capture a wide range of applicable scenarios, analyses are carried out for three distinct scenarios: (1) The retailer does not allow customer returns; (2) The

retailer allows customers to return items, and she handles the returned product herself; (3) The retailer contracts with a manufacturer and pays an additional fee to dispatch the returned products to him. The optimal return strategy and corresponding channel pricing are carried out by comparing the maximum profits using the three scenarios. This study aims to address the following questions: (1) What are the implications of the return strategies for dual-channel retailer pricing? (2) What are the benefits (detriments) of retailers contracting with manufacturers to process the returned product? This paper assumes that demand is boosted when retailers offer returns to customers. According to the study of the relevant dual-channel literature, customer returns affect pricing, and system performance are understudied. This is the first study to undertake a systematic and analytical study of contracts between retailers and manufacturers to handle returned products.

This study makes two principal contributions. First, for two-channel retailers, whether the retailer provides a return strategy depends on return rates. When the return probability is small, the retailers should provide customers with return services to enlarge the demand for the product and maximize profits. However, when the return probability is large, the retailer should cancel the return service to gain more benefit by minimizing the return losses. On this basis, the corresponding optimal pricing in two channels is proposed.

Second, the results demonstrate whether or not the offline and online retailers have contact with manufacturers, which is dependent upon the customers' sensitivity factors for pricing and return rates of products when the retailer need to provide return service to customers. If the customer's sensitivity factor for pricing is minor and the return rate is low (resp. high), the manufacturer prefers (resp. dislikes) the manufacturer to recycle the returned products to achieve maximum profit, but the retailer disliked (resp. likes) the contract. If the customers are more sensitive to the price (i.e., the demand scale is small), and if the return rate is low, both retailer and manufacturer likely achieves their maximum profits based on the wholesale market. However, if the return rate is high, the retailer would likely sign a contract to reduce return losses; the manufacturer also makes his profit based on the extra revenue made when simultaneously recycling returned products.

The rest of the paper was organized as follows. The relevant literature is reviewed in Sect. 2. The third part of this study constructs three profit functions for offline channel retailers: (1) does not provide the return service to customers, (2) offers the return service without contracts, (3) holds a return contract with manufacturers and then models the manufacturer's profit. On this basis, the optimal results are analyzed and compared, and the corresponding management insights are proposed. Section 4 focuses on the scenario of the online channel. Section 5 compares the optimal strategies for the two channels under different return probabilities. In Sect. 6, we verify the proposed results based on synthesis data. Section 7 summarizes the primary points of this paper and offers suggestions for future research.

## 2 Literature review

This section primarily reviewed two streams: channel pricing and return rate and return policy.

### 2.1 Channel pricing and return rate

Keteaberg and Zuidijk (2009) reported that demand was related to price and market size; factors such as return ratio, return volatility, and return strategy played a key role in affecting the demand for products. Su (2009) pointed out that consumers' returns may affect incentives, and they proposed a supply chain coordination strategy for consumer returns. Shuliman et al. (2011) studied return when there was market competition without considered product reference information for consumers. Liu et al. (2012) proposed relationships between pricing strategy, consumer returns, and product modularity. Balakrishnan et al. (2014) believed that if consumers browsed online first and then bought offline, it reduced the return. However, it increased the competition between the two channels and led to simultaneously profit reduction. Xu et al. (2015) studied the impacts of return prices and return period on consumers' return and retailers' inventory strategies. Sahoo et al. (2018) reported that product reviews online affect product uncertainty and affect product return rates. Li and Jiang (2019) established a model that involved a risk-neutral supplier and a risk-averse retailer, and both channels provided return services.

Considering the return rates, Gao and Su (2017) studied the optimal physical inventory for omnichannel in which they fixed the probability of a return and the price parameters. Nageswaran et al. (2020) recognized the return rate was relative to the return channel decision. Hu et al. (2018) built a product return model and proposed the seller's benchmark pricing policy. They also proved that although return probability had a relationship with price, it did not impact the model and the policy. Similarly, Gao et al. (2021) assumed that return rates related to the type of products and assume that the return rate is different for different type of products. Following the above research, in this study, it is assumed that the return rate corresponding to the products' type and two channels had different probabilities.

### 2.2 Return policy

Shulman et al. (2010) found that the return penalty may be more severe when returns are salvaged by a channel member who derives higher value from a returned unit. Choi (2013) examined the optimal return service charge policy for a fashion mass customization program. Ülkü et al. (2013) found that a retailer operating in an environment where consumers were not sensitive to return policies, and they should be cautious when setting the return timeframe. Gümüs et al. (2013) characterized the necessary and sufficient conditions under which a returns contract involved equilibrium strategy and the corresponding channel decisions. Altug and Aydinliyim

(2016) studied how consumer discount-seeking purchases affect online retailers' return policy choices. Moon et al. (2010) presented a continuous-time optimization model for a dynamic pricing and inventory control problem in a dual-channel supply chain system. Hu et al. (2018) found that ignoring returns led to overpricing and caused much loss when the demand is high.

Guo (2009) proposed that partial refunds may shift the nature of strategic interactions between service providers from partial monopolies to a competitive mechanism, thus benefitting by adjusting the efficiency of partial refunds. Liu et al. (2014) determined that when the refund is caused by the customer's return following an exogenous variable, the repurchase contract could not generate the supply chain's coordination. Yoo (2014) stated that improving product quality must precede the making of generous return policies. Tran et al. (2018) revealed that the manufacturer faced high profit variations under the quota policy; however, the variance in profits for the reseller was high under the partial-refund policy. Khouja et al. (2019) analyzed the impacts of the return policy period on regular discount retailers and occasional discount retailers, he studied the effects of providing price adjustment protection policies and their optimal deadlines and compared the cash refund policies with non-cash refund policies. He et al. (2019) developed a newsvendor model of a retailer selling his products to consumers through online and offline channels, where consumers can return unsatisfactory products to the retailer for a full refund, and the returned items can be resold as refurbished products. Samorani et al. (2019) argued that a customer who returns a product to the store might buy another product, which showed how retailer profits can benefit from allowing returns. Nageswaran et al. (2020) examined cross-channel returns; they determined that a full-refund policy increased product demand.

Unlike the studies described, this paper examines the impact of the contract between the retailer and the manufacturer on the retailers' profits and corresponding optimal channel pricings. Based on previous findings (Gao and Su 2017; Samorani et al. 2019; Nageswaran et al. 2020), this study assumes that a full-refund policy would increase product demand.

## 3 Offline channel pricing and optimal return strategy

This paper follows the subsequent modeling premise: first, retailers' and manufacturers' behavioral preferences are not considered, so the retailer and the manufacturer are rational and pursue maximum profits. Second, the loss differences between quality and non-quality issues and the resale price of returned products are not considered.

It is assumed that the total potential size of the market for one product with two channels online and offline is A, and in such a setting, the competition between offline and online channels is represented by parameters $\gamma$ and $1-\gamma$. For the offline channel, $\gamma$A represents the market scale of offline channel. Concerning consumers' sensitivity for pricing, following the previous study, let b and k show the sensitivity factors when retailers do not provide return services and do provide the full refund return service, respectively. Without loss of generality, there is $b > k > 0$, which

also implied that providing return service can promote the demand for the products (Nageswaran et al. 2020).

From the retailers' perspectives, for the offline channel, it is true that the demand function is $D_s$ and the corresponding retailing price is $P_s$. Thus, the offline demand function is relative to the total market demand, the competition parameter, the retailing price, and the consumers' sensitivity parameter. The set up allowed $C_s$ to represent the marginal cost per unit for offline channel product, and $\pi_s$ denotes the profit. Similarly, following the previous work (Gao and Su 2016; Gao et al. 2021), customers' return percentage using the offline channel $r_s$ is related to the types of products. Therefore, retailers' profits and the optimal solutions achieved using these defined parameters.

### 3.1 Decision analysis without return service

In this section, the optimal pricing and profit when offline channel retailers do not adopt a return service for consumers are explored. With regard to previous research (Padmanabhan and Png 1995; Cui et al. 2007), it is assumed that demand and price are inversely proportional. Based on Sect. 3, it is known that the total market demand is A, and the market demand segmentation proportion of the offline channel is γ. When the retailer does not use a return strategy, the customer's sensitivity coefficient factor of the price is b, so the offline channel's demand function is:

$$D_s = \gamma A - bP_s \tag{1}$$

Then, the offline profit function is:

$$\pi_s = \left(P_s - C_s\right)D_s \tag{2}$$

Therefore, the following optimal results are yielded in Lemma 1.

**Lemma 1** *When retailers do not adopt a return policy for the customers, the optimal pricing and optimal profit for the offline channel is:*

$$\begin{cases} P_s^* = \frac{\gamma A + bC_s}{2b} \\ \pi_s^* = \frac{\left(\gamma A - bC_s\right)^2}{4b} \end{cases} \tag{3}$$

With intensified market competition, to improve customer satisfaction and increase product sales, offline retailers provide additional return service to the customers to promote service level. Next, the study provides an analysis of how the offline retailers adopt return service for customers.

### 3.2 Decision analysis of retailer

To show the benefit of the contract for the retailer, a model is built that shows a retailer providing return service but not contracting with a manufacturer. Then, a situation including a retailer contact with a manufacturer to handle the returned product is targeted.

#### 3.2.1 Model setup of retailer without contract

This study assumes that offering the return service stimulates customers' purchase desires, so the market demand for products is enhanced. Following the previous research (Nageswaran et al. 2020), it is also assumed that when the retailers provided the return service, the product sales is increased by adjusting the coefficient of the demand function (i.e.,$b > k > 0$).The share of the segmentation demand of the offline market is still $\gamma$, and when the retailer give the customer a return service. In regard to previous studies (Padmanabhan and Png 1995; Cui et al. 2007), the demand function of the offline channel is shown in Eq. (4):

$$D_{s1} = \gamma A - kP_{s1}. \tag{4}$$

In this paper, $P_{s1}$ denotes the offline channel price when a retailer and a manufacturer do not reach a contract, and $D_{s1}$ indicates the corresponding product demand. It is assumed that the salvage value of the per-unit returned products is S. Thus, the profit function of the offline seller is:

$$\pi_{s1} = (P_{s1} - C_s)(1 - r_s)D_{s1} - (C_s - S)r_sD_{s1} \tag{5}$$

where $\pi_{s1}$ indicates retailers' offline profits, $(C_s - S)r_sD_{s1}$ indicates the loss of retailers due to returned products, $P_{s1} - C_s$ indicates the profits per unit of retailers, and $(1 - r_s)D_{s1}$ indicates the number of sold items. Therefore, the following optimal results in Lemma 2 are resultant.

**Lemma 2** *When retailers adopt a return strategy for the customers, the optimal pricing and optimal profit for the offline channel can be found, as shown in Eq.* (6):

$$\begin{cases} P_{sl}^* = \frac{\gamma A(1-r_s)+k(C_s-Sr_s)}{2k(1-r_s)} \\ \pi_{sl}^* = \frac{[\gamma A(1-r_s)-k(C_s-Sr_s)]^2}{4k(1-r_s)} \end{cases}. \tag{6}$$

Next, the optimal results in Eq. (6) are analyzed with the return rate. Here, $r_s < \left(\gamma A - C_s k\right)\big/(\gamma A - Sk) = \bar{r}_s$ (e.g., $D_{s1} > 0$). It is intuitive that if one type of products' return rate is high, the retailer would choose not to sell it anymore to avoid return losses.

### 3.2.2 Property analysis

To analyze the impact of return rate on optimal pricing and optimal profit, respectively, $P_{s1}^{*}$ and $\pi_{s1}^{*}$ were used separately to derive the return rates by getting the first-order derivatives.

The following was yielded:$dP_{s1}^{*}/dr_s = \left(C_s - S\right)/2(1 - r_s)^2 > 0$. With the increased return rates, the optimal price was higher. Therefore, when retailers provide the return service to customers, they can only offset the return loss by increasing the product's price, thus the larger the return probability, the higher the product pricing.

The following is true: $\frac{d\pi_{s1}^{*}}{dr_s} = [\gamma A(1 - r_s) - k(C_s - Sr_s)]\frac{-k(C_s - S) - [\gamma A - kS](1 - r_s)}{4k(1 - r_s)^2} < 0$; therefore, the optimal profit decreased with the return rate increased. There were two reasons: first, the price rose when the return rate increased. Although the unit product's profit grew with the increasing optimal pricing, the demand declined with the pricing increases, and the resulting return losses expanded. Thus, optimal profit shrank with the increase in return rate.

To summarize, even if offering a return policy stimulates sales, with the increased return rate, the negative effect of customer returns on profits was greater than the positive effects. According to the demand function adopted in this paper, retailers should adopt to lower the price of products and increase sales to get maximum profits.

To reduce the loss of the customers' higher return probability and make more profit, the situation where a retailer pays an extra fee and contracts with a manufacturer to handle the returned products will be analyzed in the next section.

### 3.2.3 Model setup with contract

With increase customer return rates, then return is costly, retailers cannot manage the return products. It may be suboptimal for the retailer to sell a product when bearing the cost of such returns. It would be suitable for the retailer to ask the manufacturer to recycle the returned product. In this paper, retailers who contract with the manufacturer by paying an extra fee (i.e., when the retailer purchases the goods, the manufacturers charge a certain fee to each product above the wholesale price) are studied. Furthermore, to assess the implication of the contract, it is adopted that the additional retailer cost for the contract is $W_s$; however, this generates extra revenue for the manufacturer. In agreement with the assumptions concerning the contract, the retailer can dispatch the returned products to the manufacturer and transfer return losses to the manufacturers. The offline channel retailer's product price is $P_{S2}$ when the retailers make a contract with the manufacturer. Aligning with previous studies (Padmanabhan and Png 1995; Cui et al. 2007), the demand function is:

$$D_{s2} = \gamma A - kP_{s2}. \tag{7}$$

Moreover, the offline profit function is:

$$\pi_{s2} = (P_{s2} - C_s - W_s)(1 - r_s)D_{s2} \tag{8}$$

Here, $W_s$ denotes the unit cost of offline retailers increased when the retailers contract with the manufacturers. In this way, the retailer can transfer the return loss (i.e., $(C_s - S)r_sD_{s1}$ in Eq. (5)) to the manufacturers under this contract. Here, the $(1 - r_s)D_{s2}$ in Eq. (8) indicates the demand for products, and the retailer's optimal results, as shown in Eq. (9).

**Lemma 3** *The optimal pricing and optimal profit are shown in Eq.* (9)*, and it indicates when the retailers reach a contract with the manufacturers*:

$$\begin{cases} P_{s2}^* = \frac{\gamma A + (C_s + W_s)k}{2k} \\ \pi_{s2}^* = \frac{[\gamma A - k(C_s + W_s)]^2}{4k}(1 - r_s) \end{cases} \tag{9}$$

Next, the characters for the optimal results in Eq. (9) are analyzed with the return rate. Here, $k < \gamma A/(C_s + W_s) = \bar{k}_s$ (e.g., $D_{s2} > 0$). Clearly, consumers' sensitivity factors for pricing is large, and the retailer also wants to pay extra money to the manufacturer to handle the returned products. It may be a better choice for the retailer to stop selling this product.

### 3.2.4 Property analysis

To analyze the impacts of return rates on optimal pricing and optimal profit, $P_{s2}^*$, and $\pi_{s2}^*$ were used separately to derive $r_s$ and obtain the following:

Due to the optimal pricing, $P_{s2}^*$ is unrelated to the return rate, which indicate that customers' returns did not affect the pricing. The reason was when a contract between retailer and manufacturer was filed, the return loss transferred from the retailer to the manufacturer. However, the retailer had to bear higher unit cost in this situation.

The following is true: $d\pi_{s2}^*/dr_s = -[\gamma A - k(C_s + W_s)]^2/4k < 0$, which means the optimal profit monotonically decreased with the increase in return rate. According to the previous analysis in Sect. 3.2.2, the return rate affected retailers' profits. When other parameters (except offline return rate) were stationary, retailers should take advantage of the offline store to increase optimal profits by reducing the return rates through touch and feel.

When a retailer contracts with a manufacturer, the manufacturer should be the decision-maker, meaning the manufacturer should gain more profit with a contract than without a contract. The following section models the objective function of the manufacturer.

## 3.3 Decision analysis of manufacturer

Section 3.2 discussed the implication of the contract between the manufacturers and retailer, and it was assumed that the retailer paid an additional cost to the manufacturers, which was extra revenue for the manufacturer. Following that assumption, if

the offline channel retailer contracts with the manufacturer by paying an extra fee, the retailer mails the returned products to the manufacturer. Without loss of generality, both retailer and manufacturer have the same salvage value S for each returned product. Similarly, to examine the benefit of the contract for the manufacturer, a model depicting a retailer without contacts and one with contracts with a manufacturer to handle the returned products is needed.

The marginal cost of each product for the manufacturer is C. Without loss of generality, $C_s > C > S$. To simplify, the manufacturers' production is equivalent to the retailers' sales (i.e., customers' demand). The optimal results in Lemmas 2 and 3 are employed and combined with the demand functions in Eqs. (4) and (7). The profit-function for the manufacturer without a contract is:

$$\pi_{s1}^{m*} = (C_s - C)D_{s1}^* = (C_s - C)\frac{\gamma A(1 - r_s) - k(C_s - Sr_s)}{2(1 - r_s)}. \tag{10}$$

Here, $C_s - C$ represents the manufacturer's profit for each product without a contract with offline retailers. Note that $D_{s1}^*$ is the total sales/demand of the product when the retailer provides return service to customers.

Thus, the profit function of manufacturer with contract is:

$$\begin{aligned}\pi_{s2}^{m*} &= (C_s + W_s - C)(1 - r_s)D_{s2}^* - (C - S)r_s D_{s2}^* \\ &= \left[(C_s + W_s - C)(1 - r_s) - (C - S)r_s\right]\frac{\gamma A - (C_s + W_s)k}{2}\end{aligned} \tag{11}$$

where $C_s + W_s - C$ represents the per-unit profit of the manufacturer for online retailers with a contract. The total sales/demand of the product when the retailer provides return service to customers is $D_{s2}^*$. The cost of the manufacturer for each returned product is denoted by $C - S$.

### 3.4 Analysis of optimal solutions and optimal strategy

In Sects. 3.1–3.3, the identical optimal profits were investigated for different situations. This section compared the optimal solutions and examined whether or not offline retailers should provide return services to customers and whether or not the retailers should have a contract with manufacturers, in addition to when a contract should be approached by both sides.

According to Eq. (3), the optimal profit for an offline channel retailer is $\pi_s^*$ when the offline channel retailer does not provide the return service; the optimal profit for an offline channel retailer is $\pi_{sl}^*$ when there is no contract between the offline retailer and the manufacturer, as shown in Eq. (6). By comparing the retailers' optimal profits in these two cases, the following results were yielded:

**Proposition 1A** (1) *If* $0 < r_s \le \hat{r}_s$, *there is* $\pi_s^* \le \pi_{s1}^*$;

(2) *If* $\hat{r}_s < r_s < \bar{r}_s$, *there is* $\pi_s^* > \pi_{s1}^*$.

$$where, \hat{r}_s = \frac{\left[2b(\gamma A - kC_s)(\gamma A - kS) - k\left(\gamma A - bC_s\right)^2\right] - \sqrt{\left[4kb(\gamma A - kS)\left(\gamma A - bC_s\right)^2 k(C_s - S) + k^2\left(\gamma A - bC_s\right)^4\right]}}{2b(\gamma A - kS)^2}.$$

Proposition 1A shows that when the return probability is small, the retailer should provide the return service to achieve maximum profit. Although the returned products generate losses, return service can stimulate a customer's consumption and increase a product's sales to offset the return losses and enhance the profits. When the return rate is high, the retailer should cancel the return service to reduce the losses and get optimal profits. The retailer needs to trade-off the profit earned with increased demand and the loss from returns. However, this recommendation seems infeasible and not grounded in online markets' realities or in consumers' expectations for returning products. This study suggests that a contract between manufacturers and retailers help retailers maximize their profits.

In Eq. (6), the optimal profit of the offline channel retailer is $\pi_{sl}^{*}$ when there was no contract between the offline retailer and the manufacturer. Moreover, the offline channel retailer's optimal profit is $\pi_{s2}^{*}$ when there is a contract, as shown in Eq. (9). By comparing the retailer's optimal profits $\pi_{sl}^{*}$ and $\pi_{s2}^{*}$, the results are:

**Proposition 1B** (1) *If there is* $r_s \le W_s/(C_s + W_s - S) = r_{s1}$, *we will get* $\pi_{sl}^{*} \ge \pi_{s2}^{*}$;

(2) *If there is* $W_s/(C_s + W_s - S) = r_{s1} < r_s < \bar{r}_s$, *we will get* $\pi_{sl}^{*} \ge \pi_{s2}^{*}$.

Proposition 1 B is straightforward. If the offline channel retailer provides the return service to the customer, then the retailer can manage the returned products when the return rate is low. The retailer did not require shifting the return loss by paying extra costs to sign a contract. However, the retailer needed to consume an additional cost to make the contract, as mentioned above, with the increased return rates, and then the considerable return loss was transferred to the manufacturer, thereby increasing the retailer's total optimal profit.

In Eq. (10), the optimal profit of the manufacturer is $\pi_{s1}^{m*}$ when there was not a contract between the offline retailer and the manufacturer. The optimal profit of the manufacturer is $\pi_{s2}^{m*}$ when there was a contract in Eq. (11). By comparing the manufacturer's optimal profits $\pi_{s1}^{m*}$ and $\pi_{s2}^{m*}$, this was the conclusion:

**Proposition 1C**

(1) *When* $0 < k < \bar{k}_{s1}$ *holds*,

(1) *If there is* $0 < r_s < r_{s1}$, *we will get* $\pi_{s1}^{m*} < \pi_{s2}^{m*}$;
(2) *If there is* $r_{s1} \le r_s \le \bar{r}_s$, *we will get* $\pi_{s1}^{m*} \ge \pi_{s2}^{m*}$.

(2) *When* $\bar{k}_{s1} < k < \bar{k}_{s2}$ *holds*,

(1) *If there has* $r_{s1} \le r_s \le r_{s3}$, *we will get* $\pi_{s1}^{m*} \ge \pi_{s2}^{m*}$;

(2) *If there has* $r_s < r_{s1}$ *or*, $\bar{r}_s \geq r_s > r_{s3}$, *we will get* $\pi_{s1}^{m*} < \pi_{s2}^{m*}$.

$$Where, \begin{cases} \bar{k}_{s1} = (C - S)\gamma A/((C_s - C)(C_s + W_s - S) + (C_s + W_s)(C - S)) \\ \bar{k}_{s2} = \gamma A(C_s - S)/[(C_s + W_s)(C_s - S) + (C_s - C)(C_s - S + W_s)]. \\ r_{s3} = [(\gamma A - k(C_s + W_s)) - k(C_s - C)]/[\gamma A - k(C_s + W_s)] \end{cases}$$

Unlike propositions 1A and 1B, the consumers' sensitivity factor $k$ for pricing played a key role in Proposition 1C. If the sensitivity was low, which meant the customers' demands were high, the results were similar—the manufacturer preferred (resp. does like) a contract with the retailers when the return rate was low (resp. high). From the manufacturer's perspective, the extra revenue (resp. cannot) offset the cost to process the returned product when the return rate was low (resp. high). If the consumers' sensitivity factor for pricing was high, it indicated the product's sales were small. The most prominent finding was that when the return rate was high, the manufacturer preferred to contract with retailers to maximize profit. Considering when the demand was low, the manufacturer could not get high profits through the traditional wholesale market, and he will choose to earn more profit by accepting extra revenue from the retailer for recycling the returned products.

This section did not consider the relationship between the return probability and the extra cost in the negotiation for the contract because the manufacturer may not be able to obtain the accurate return probability. Based on the above propositions, the following results were found:

**Proposition 2**

(1) *When* $0 < k < \bar{k}_{s1}$ *(i.e.,* $0 \leq r_{s1} \leq \bar{r}_s \leq r_{s3}$*) holds*,

(a) *If* $r_{s1} \leq r_s \leq \bar{r}_s$, *there are* $\pi_{s1}^{m*} \geq \pi_{s2}^{m*}$ *and* $\pi_{s1}^{*} < \pi_{s2}^{*}$ *(i.e., retailer likes but manufacturer dislikes contract)*;

(b) *If* $r_s < r_{s1}$, *there are* $\pi_{s1}^{m*} < \pi_{s2}^{m*}$ *and* $\pi_{s1}^{*} \geq \pi_{s2}^{*}$ *(i.e., retailer dislikes but manufacturer likes contract).*

(2) *When* $\bar{k}_{s1} \leq k < \bar{k}_{s2}$ *(i.e.,* $0 \leq r_{s1} \leq r_{s3} \leq \bar{r}_s$*) holds*,

(c) *If* $0 < r_s \leq r_{s1}$, *there are* $\pi_{s1}^{m*} \geq \pi_{s2}^{m*}$ *and* $\pi_{s1}^{*} < \pi_{s2}^{*}$ *(i.e., retailer likes but manufacturer dislikes contract)*;

(d) *If* $r_{s1} < r_s \leq r_{s3}$, *there are* $\pi_{s1}^{m*} < \pi_{s2}^{m*}$ *and* $\pi_{s1}^{*} \geq \pi_{s2}^{*}$ *(i.e., retailer dislikes but manufacturer likes contract)*;

(e) *If* $r_{s3} < r_s < \bar{r}_s$, *there are* $\pi_{s1}^{m*} < \pi_{s2}^{m*}$ *and* $\pi_{s1}^{*} < \pi_{s2}^{*}$ *(i.e., both retailer and manufacturer like contract).*

(3) (*When* $\bar{k}_{s2} \leq k < \bar{k}_{s3}$ *(i.e.,* $0 \leq r_{s3} \leq r_{s1} \leq \bar{r}_s$*) holds*,

(f) *If* $0 < r_s \le r_{s3}$, *there are* $\pi_{s1}^{m*} < \pi_{s2}^{m*}$ *and* $\pi_{s1}^{*} \ge \pi_{s2}^{*}$ *(i.e., retailer dislikes but manufacturer likes contract)*;

(g) *If* $r_{s3} < r_s \le r_{s1}$, *there are* $\pi_{s1}^{m*} \ge \pi_{s2}^{m*}$ *and* $\pi_{s1}^{*} \ge \pi_{s2}^{*}$ *(i.e., both retailer and manufacturer dislike contract)*;

(h) *If* $r_{s1} < r_s < \bar{r}_s$, *there are* $\pi_{s1}^{m*} < \pi_{s2}^{m*}$ *and* $\pi_{s1}^{*} < \pi_{s2}^{*}$ *(i.e., both retailer and manufacturer like contract).*

(4) *(When* $\bar{k}_{s3} \le k < \bar{k}_s$ *(i.e.,* $r_{s3} \le 0$ *and* $\bar{r}_s > r_{s1} \ge 0$*) holds,*

(i) *If* $0 < r_s \le r_{s1}$, *there are* $\pi_{s1}^{m*} \ge \pi_{s2}^{m*}$ *and* $\pi_{s1}^{*} \ge \pi_{s2}^{*}$ *(i.e., both retailer and manufacturer dislike contract)*;

(j) *If* $r_{s1} < r_s < \bar{r}_s$, *there are* $\pi_{s1}^{m*} < \pi_{s2}^{m*}$ *and* $\pi_{s1}^{*} < \pi_{s2}^{*}$ *(i.e., both retailer and manufacturer like contract).*

*Where,* $\bar{k}_{s3} = \gamma A/(2C_s + W_s - C)$ *and* $0 \le \bar{k}_{s1} \le \bar{k}_{s2} \le \bar{k}_{s3} \le \bar{k}_s$.

Proposition 2 shows the beneficial conditions for which offline channel retailers and manufacturers like to contract to handle returned products. Thus, the offline channel retailer preferred to pay an extra fee on each product to the manufacturer to avoid return losses when the return rate was high, and the manufacturer also achieved more profit via the extra revenue to recycle the returned products when the consumers' sensitivity factor for pricing was high, and the demand was low. From Proposition 2, the conditions were found for an offline channel retailer and a manufacturer to dislike contracts for handling the returned products, and it defined conditions for which only one side was agreeable to the contract.

In summation, the results showed that when customers' sensitivity factor for pricing is high (i.e., the demand is small), and the return rate is low, both retailer and manufacturer dislike contracting to handle the returned products. However, when both the customer's sensitivity factor for pricing and the return rate are high, both retailer and manufacturer like to contract to handle the returned products. Otherwise, only one side preferred to contract with another. Commonly, when customers' sensitivity factor for pricing was minor, or the demand was large, and the return rate was low, the manufacturer preferred the retailer to pay an extra fee to contract and achieve maximum profit. However, the retailer disliked the contract because of the additional contract cost. If the return rate was high, the manufacturer disliked the retailer paying an extra fee to avoid the cost of product recycling; however, the retailer liked to sign the contract to reduce the return losses. Based on the above conclusions, the following management insights can be drawn:

***Managerial insights 1:*** *If the retailers provided return services to customers, the retailers could control the returned goods when the return rate was low. When the return rate was high, and the consumers' sensitivity factor for pricing was high, the retailer should sign a contract to offset the return losses to maximize the profit, and the manufacturer also achieved higher profit based on the extra fee to recycle the returned products. However, when the sales were small and the return rate was low, the retailer did not enjoy paying extra fees to handle the returned product. The*

*manufacturer's revenue to recycle the returned product could not compensate for his losses.*

## 4 Online channel pricing and optimal return strategy

This section focuses on the model and optimal solution for retailers who use online channel. Following the assumptions made in Sect. 3 for offline channel retailers, the market's total size for the corresponding product was defined. Here, $(1-\gamma)\cdot A$ represents the market scale of the online channel. For simplicity of comparison regarding the online channel, b and k were adopted separately to represent the customers' pricing sensitivity coefficients when the retailers did not provide and did provide return policies in the market, where $b > k > 0$.

In contrast to the offline channel, for the online channel's demand function is $D_0$, the retailing price is $P_0$, and the marginal cost per unit offline channel product is $C_0$. The terms $r_0$ and $W_0$ are used to denote the online return rate and additional costs from the contract with the manufacturer. Without loss of generality, $0 \le r_s \le r_0 \le 1$. The additional cost to be paid to the manufacturer under two channels is different, and without loss of generality, there is $W_s \le W_0$ (i.e., the additional cost increases with the increasing return rate). The returned products have the same salvage value whether they are returned by online channel customers or offline customers (i.e., $C_s > C_0 > S$).

### 4.1 Decision analysis without return service for online channel

Following the same assumption outlined in Sect. 3, the online channel's demand function is the same as the offline channel. The difference between the online and the offline is the segmentation ratio of the market demand for the online channel,$1-\gamma$. The proportion of the two-channel market segmentation affects the retailers' pricing and return strategies. However, customers' preferences and the market distribution proportions between the two channels were not considered in this paper; it is a topic for future research.

When the retailer does not provide a return service to the customer, its sensitivity coefficient for pricing is b. The demand function of the offline channel is:

$$D_0 = (1-\gamma)A - bP_0 \tag{12}$$

where, $D_0$ represents the total demand of online sellers, and the ratio of $1-\gamma$ represents the market segmentation of online channels. The profit function of the online channel retailer is:

$$\pi_0 = \left(P_0 - C_0\right)D_0. \tag{13}$$

Similarly, function $\pi_0$ is optimized, and the corresponding optimal results are obtained.

**Lemma 4** *The optimal pricing and profit for online channel retailers when retailers do not adopt return service to the customers are*:

$$\begin{cases} P_0^* = \frac{(1-\gamma)A+bC_0}{2b} \\ \pi_0^* = \frac{\left((1-\gamma)A-bC_0\right)^2}{4b} \end{cases}. \tag{14}$$

As competition intensified, online retailers actively adopt new technologies to improve customers' perceptions of product quality and introduce return services to reduce customers' concerns and stimulate product sales. Next, a situation when an online retailer provides a return service to customers but does not have a contract with the manufacturer is analyzed.

### 4.2 Decision analysis without contract for online channel

To show the benefit of the contract for the online channel retailer, this study presents a model when the retailer provided a return service without the contract between retailer and manufacturer. Then, focus is given to when the retailer contacts the manufacturer to handle the returned products.

#### 4.2.1 Model setup and analysis

As previously mentioned in Sect. 3, we did not consider the change of the segmentation proportion fir market demand between two channels when the retailers adopted the return service for the customer. However, it was assumed that the demand was boosted when retailers adopted the return service. The parameters for customers' sensitivity on pricing was changed to indicate $b > k > 0$.

The uncertainty of product quality affects the willingness of customers to purchase online. In this situation, if the online retailers provide the return service to customers, the online purchase improved, thereby affecting the segmentation ratio of the dual-channel market. This is a future potential research direction.

Regarding above assumptions, the total market demand is A, the demand segmentation ratio of the online market is $1-\gamma$, and when the retailer allows returned products, the consumer's sensitivity coefficient to pricing declined to k, so the demand function of the online channel is:

$$D_{01} = (1-\gamma)A - kP_{01}. \tag{15}$$

Thus, the profit function of the online seller is:

$$\pi_{01} = (P_{01} - C_0)(1-r_0)D_{01} - (C_0 - S)r_0D_{01}. \tag{16}$$

The demand function Eq. (15) shows that the primary difference between the online seller's and offline seller's profit functions are the market segmentation ratios. Similarly, when there is no contract between online retailers and manufacturers, the corresponding optimal results are:

**Lemma 5** *The optimal pricing and profit for the online channel when the retailer and the manufacturer do not make a contract is*:

$$\begin{cases} P_{01}^{*} = \frac{(1-\gamma)A(1-r_0)+k(C_0-Sr_0)}{2k(1-r_0)} \\ \pi_{01}^{*} = \frac{[(1-\gamma)A(1-r_0)-k(C_0-Sr_0)]^2}{4k(1-r_0)} \end{cases} \tag{17}$$

Next, this study analyzes the characters of the optimal results in Eq. (17) with the return rate considered. Similarly, as in Sect. 3.2.1, here, $r_0 < ((1-\gamma)A - C_0k)/((1-\gamma)A - Sk) = \bar{r}_0$ (e.g., $D_{01} > 0$).

#### 4.2.2 Property analysis

To analyze the impact of return rate on optimal results, $P_{01}^{*}$ and $\pi_{01}^{*}$ were used to conduct the first order derivation on online return rates $r_0$ separately, and that yielded: $dP_{01}^{*}/dr_0 > 0$ and $d\pi_{01}^{*}/dr_0 < 0$.

Therefore, for online retailers regarding pricing and profit, there were similar conclusions to those for offline retailers. With the increased probability of return, a higher commodity price was required. Although the profit per product unit increased, the product demand decreased in the face of increased optimal pricing. As the return probability increased, the retailer's return losses increased. Meanwhile, the optimal profit decreased with the increased return rate.

#### 4.2.3 Model setup with contract of online channel retailer

When there is contract between an online retailer and a manufacturer, the price of the online channel retailer is $P_{02}$, and then the demand function is:

$$D_{02} = (1-\gamma)A - kP_{02}. \tag{18}$$

The profit function of the online channel is shown in Eq. (19):

$$\pi_{02} = (P_{02} - C_0 - W_0)(1-r_0)D_{02}, \tag{19}$$

where $W_0$ represents the extra cost per unit cost charged by the manufacturer for online retailers to make a contract. By optimizing the function of $\pi_{02}$, the optimal results are obtained in Eq. (20).

**Lemma 6** *The online retailer's optimal pricing and optimal profit are*:

$$\begin{cases} P_{02}^{*} = \frac{(1-\gamma)A+(C_0+W_0)k}{2k} \\ \pi_{02}^{*} = \frac{[(1-\gamma)A-k(C_0+W_0)]^2}{4k}(1-r_0) \end{cases}. \tag{20}$$

Similarly, here, we also have $k < (1-\gamma)A/(C_0+W_0) = \bar{k}_0$ (e.g., $D_{02} > 0$).

#### 4.2.4 Property Analysis for Online Channel Retailer

To analyze the impact of return rate on optimal pricing and optimal profit, $P_{s1}^{*}$ and $\pi_{s1}^{*}$ were used separately to derive the first order of the online return rate for $r_0$.

(1) The optimal pricing $P_{02}^{*}$ is unrelated to the return rate, meaning that the return rate did not affect optimal pricing. The main reason was that the online retailer transferred the return losses to the manufacturer by using the contract. Thus, the optimal pricing no longer included the return probability; however, the retailer had to bear a higher unit cost (i.e., $W_0 + C_0$).
(2) According to Eq. (20), the optimal profit, still strictly decreased with the increased return rate. Generally, the difference between offline retailers and online retailers is that online products have higher uncertainty levels and greater customer return rates. Therefore, online sellers should try to improve customer perception and satisfaction with the product through online high-tech, and they should provide detailed information and comments for customers to reduce the return rates.

In Sect. 4.3, the optimal strategy for contracting from the perspective of manufacturer is analyzed.

### 4.3 Decision analysis of manufacturer

In agreement with Sect. 3.3, the manufacturer's cost is C. Without loss of generality, $C_o > C > S$. By employing the optimal results in Lemmas 5 and 6 and combining the demand functions for online channel in Eqs. (15) and (18), the profit function for a manufacturer without a contract is:

$$\pi_{o1}^{m*} = (C_o - C)D_{o1}^{*} = (C_o - C)\left(\frac{(1-\gamma)A}{2} - \frac{k(C_0 - Sr_0)}{2(1-r_0)}\right) \tag{21}$$

$C_o - C$ represents the manufacturer's profit for each product and without a contract for online retailers. Note that $D_{o1}^{*}$ is the total sales/demand of the product when the retailer provided return service to customers.

Thus, the profit function of manufacturer with contract is:

$$\pi_{o2}^{m*} = (C_o + W_o - C)(1 - r_o)D_{o2}^{*} - (C - S)r_o D_{o2}^{*} = \left[(C_o + W_o - C)(1 - r_o) - (C - S)r_o\right] \left(\frac{(1-\gamma)A}{2} - \frac{(C_0 + W_0)k}{2}\right), \tag{22}$$

where $C_o + W_o - C$ represents the per-unit profit of the manufacturer for online retailers with a contract. Note that $D_{o2}^{*}$ is the total sales/demand of the product when the retailer provides return service to customers, and $C - S$ denotes the cost of the manufacturer for each returned product.

### 4.4 Analysis of optimal solutions for online channel

The main difference between optimal pricing and profit for offline and online focused on the market segmentation proportion: offline is $\gamma$ and online $1-\gamma$.Additionally, the online return probability was larger than offline, so the online return rate had a more significant negative impact on profits than the offline return rate. However, considering the product unit costs, offline retailers usually bore higher costs than online retailers for items such as rent, inventory, and attendants, so, $C_s > C_0$ without loss of the generality.

Similarly, according to Eq. (14), the optimal profit is $\pi_0^*$ when the online retailer does not provide the return service to customers, and the optimal profit is $\pi_{01}^*$ when there is no contract between offline retailers and manufacturers in Eq. (17). Thus, the following is true:

**Proposition 3A** (1) *If there is* $r_0 \le \hat{r}_0$, *then we will obtain* $\pi_0^* \le \pi_{01}^*$;

(2) *If there is* $r_0 > \hat{r}_0$, *then we will obtain* $\pi_0^* > \pi_{01}^*$.

$$\textit{Here,} \left( \frac{\left[2b((1-\gamma)A-kC_0)((1-\gamma)A-kS)-k\left((1-\gamma)A-bC_0\right)^2\right]}{2b[(1-\gamma)A-kS]^2} - \frac{\sqrt{\left[4kb((1-\gamma)A-kS)\left((1-\gamma)A-bC_0\right)^2k(C_0-S)+k^2\left((1-\gamma)A-bC_0\right)^4\right]}}{2b[(1-\gamma)A-kS]^2} \right) = \hat{r}_0.$$

In general practice, the online channel has higher return rate than offline channel. To process with the returned products, some retailers may increase pricing. However, the higher price yields lower demand, thus reducing the optimal profit. Online retailers need to trade-off the profit brought on by the increased product sales and the returned products' losses.

According to Eq. (17), the optimal profit is $\pi_{01}^*$ when there is no contract between online retailers and manufactures. The optimal profit is $\pi_{02}^*$ when there is a contract in Eq. (20). The retailer's optimal profit in both cases are compared to get the following:

**Proposition 3B**

1) *If there is* $0 < r_0 \le W_0/(C_0 + W_0 - S) = r_{01}$, *then we will get* $\pi_{01}^* \ge \pi_{02}^*$;
2) *If there is* $W_0/(C_0 + W_0 - S) = r_{01} < r_0 < \bar{r}_0$, *then we will get* $\pi_{01}^* < \pi_{02}^*$.

The online channel retailer must balance the relationship between the additional cost of a contract and the returned products' losses.

The optimal profit is $\pi_{02}^{m*}$ (resp. $\pi_{01}^{m*}$) for manufacturers in Eq. (22) (resp. Equation (21)) when the online channel retailer and the manufacturer have (resp. does not have) a contract. By comparing the optimal results from two situations, the following relations are made:

**Proposition 3C** *When* $0 < k < \bar{k}_{01}$ *holds*,

*If there is* $0 < r_0 < r_{01}$, *we will get* $\pi_{01}^{m*} < \pi_{02}^{m*}$;

*If there is* $r_{01} \le r_0 \le \bar{r}_0$, *we will get* $\pi_{01}^{m*} \ge \pi_{02}^{m*}$.

*When* $\bar{k}_{01} < k < \bar{k}_{02}$ *holds*,

*If there has* $r_{01} \le r_0 \le r_{03}$, *we will get* $\pi_{01}^{m*} \ge \pi_{02}^{m*}$;

*If there has* $r_0 < r_{01}$ or, $\bar{r}_0 \ge r_0 > r_{03}$, *we will get* $\pi_{01}^{m*} < \pi_{02}^{m*}$.

$$\textit{where}\begin{cases}\bar{k}_{01} = (C-S)(1-\gamma)A/((C_0-C)(C_0+W_0-S)+(C_0+W_0)(C-S))\\ \bar{k}_{02} = (1-\gamma)A(C_0-S)/[(C_0+W_0)(C_0-S)+\big(C_0-C\big)\big(C_s-S+W_0\big)]\\ r_{03} = [\big((1-\gamma)A-k(C_0+W_0)\big)-k\big(C_0-C\big)]/[(1-\gamma)A-k(C_0+W_0)]\end{cases}$$

Proposition 3C also shows that the consumers' sensitivity for pricing and return rate performs a crucial function, and similar outcomes were generated in Proposition 1C. Based on Propositions 3B and 3C, the following conclusions are proposed:

**Proposition 4**

(1) *When* $0 < k < \bar{k}_{01}$ *(i.e.,* $0 \le r_{01} \le \bar{r}_0 \le r_{03}$*) holds*,
(k) *If* $r_{01} \le r_0 \le \bar{r}_0$, *there are* $\pi_{01}^{m*} \ge \pi_{02}^{m*}$ *and* $\pi_{01}^{*} < \pi_{02}^{*}$ *(retailer agrees but manufacturer disagrees contract)*;
(l) *If* $r_0 < r_{01}$, *there are* $\pi_{01}^{m*} < \pi_{02}^{m*}$ *and* $\pi_{01}^{*} \ge \pi_{02}^{*}$ *(retailer disagrees but manufacturer agrees contract)*.
(2) *When* $\bar{k}_{01} \le k < \bar{k}_{02}$ *(i.e.,* $0 \le r_{01} \le r_{03} \le \bar{r}_0$*) holds*,
(m) *If* $0 < r_0 \le r_{01}$, *there are* $\pi_{01}^{m*} \ge \pi_{02}^{m*}$ *and* $\pi_{01}^{*} < \pi_{02}^{*}$ *(retailer agrees but manufacturer disagrees contract)*;
(n) *If* $r_{01} < r_0 \le r_{03}$, *there are* $\pi_{01}^{m*} < \pi_{02}^{m*}$ *and* $\pi_{01}^{*} \ge \pi_{02}^{*}$ *(retailer disagrees but manufacture agrees contract)*;
(o) *If* $r_{03} < r_0 < \bar{r}_0$, *there are* $\pi_{01}^{m*} < \pi_{02}^{m*}$ *and* $\pi_{01}^{*} < \pi_{02}^{*}$ *(both retailer and manufacturer agree contract)*.
(3) *When* $\bar{k}_{02} \le k < \bar{k}_{03}$ *(i.e.,* $0 \le r_{03} \le r_{01} \le \bar{r}_0$*) holds*,
(p) *If* $0 < r_0 \le r_{03}$, *there are* $\pi_{01}^{m*} < \pi_{02}^{m*}$ *and* $\pi_{01}^{*} \ge \pi_{02}^{*}$ *(retailer disagrees but manufacturer agrees contract)*;
(q) *If* $r_{03} < r_0 \le r_{01}$, *there are* $\pi_{01}^{m*} \ge \pi_{02}^{m*}$ *and* $\pi_{01}^{*} \ge \pi_{02}^{*}$ *(both retailer and manufacturer disagree contract)*;
(r) *If* $r_{01} < r_0 < \bar{r}_0$, *there are* $\pi_{01}^{m*} < \pi_{02}^{m*}$ *and* $\pi_{01}^{*} < \pi_{02}^{*}$ *(both retailer and manufacturer agree contract)*.
(4) *When* $\bar{k}_{03} \le k < \bar{k}_0$ *(i.e.,* $r_{03} \le 0$ *and* $\bar{r}_0 > r_{01} \ge 0$*) holds*,
(s) *If* $0 < r_0 \le r_{01}$, *there are* $\pi_{01}^{m*} \ge \pi_{02}^{m*}$ *and* $\pi_{01}^{*} \ge \pi_{02}^{*}$ *(both retailer and manufacturer disagree contract)*;

(t) *If* $r_{01} < r_0 < \bar{r}_0$, *there are* $\pi_{01}^{m*} < \pi_{02}^{m*}$ *and* $\pi_{01}^{*} < \pi_{02}^{*}$ *(both retailer and manufacturer agree contract).*
*Where*, $\bar{k}_{03} = (1-\gamma)A/(2C_0 + W_0 - C)$, *and* $0 \le \bar{k}_{01} \le \bar{k}_{02} \le \bar{k}_{03} \le \bar{k}_0$.

Like Proposition 2, Proposition 4 demonstrates that when customer's sensitivity factor for pricing was high or the online channel demand was small, and the return rate was low, both online channels retailer and manufacturers disagreed about contracting to handle the returned products. However, when both the customer's sensitivity factor for pricing and the return rate were high, both retailer and manufacturer agreed to the returned product handling contract. That means the online channel retailer preferred paying the manufacturer an extra fee to avoid return loss when the return rate was high, and the manufacturer achieved maximum profit through the extra revenue when the consumers' sensitivity for pricing was high, and the demand was low.

When the customer's sensitivity for pricing was small, or the demand was large when the return rate was low, the manufacturer preferred for the retailer to pay an extra fee for a contract to achieve maximum profit. However, the retailer disliked the contract because of the additional cost. If the return rate was high, the manufacturer did not like the retailer contracting to avoid recycling the returned products; however, the retailer liked to sign the contract to reduce return losses. Based on the above conclusions, the following management insights are drawn:

***Managerial insight 2:*** *Due to the higher uncertainty of online channels, the retailer should pay more attention to the return rate, which is the most crucial factor affecting the retailers' profits. Similarly, like offline retailers, when the return probability is small, the online retailer should provide the customer's return service to maximize profit. Although the return generates losses, the return service can stimulate the market demand and increase the profit. When the return probability is large, the retailer should cancel the return service to the customer and reduce the loss caused by returns. With the return rate and customers' sensitivity to pricing changes, the retailer should balance the pros and cons of shifting the return losses by paying an additional fee to the manufacturer. Therefore, online retailers should provide more detailed product descriptions, reduce return rates, increase publicity, and boost market demand.*

**Table 1** Optimal policies of offline channel retailer and manufacturer

| | Retailer likes the contract | Retailer dislikes the contract |
|---|---|---|
| Manufacturer likes the contract | 1) If $\bar{k}_{s1} \le k < \bar{k}_{s2}$ and $r_{s3} < r_s < \bar{r}_s$ hold<br>2) If $\bar{k}_{s2} \le k < \bar{k}_s$ and $r_{s1} < r_s < \bar{r}_s$ hold | 1) If $0 < k < \bar{k}_{s2}$ and $0 < r_s \le r_{s1}$ hold<br>2) If $\bar{k}_{s1} \le k < \bar{k}_{s2}$ and $0 < r_s \le r_{s3}$ hold |
| Manufacturer dislikes the contract | 1) If $0 < k < \bar{k}_{s1}$ and $r_{s1} \le r_s \le \bar{r}_s$ hold<br>2) If $\bar{k}_{s1} \le k < \bar{k}_{s2}$ and $r_{s1} < r_s \le r_{s3}$ hold | 1) If $\bar{k}_{s2} \le k < \bar{k}_{s3}$ and $r_{s3} < r_s \le r_{s1}$ hold<br>2) If $\bar{k}_{s3} \le k < \bar{k}_s$ and $0 < r_s \le r_{s1}$ hold |

## 5 Optimal pricing and strategies for the dual channels

This section analyzes optimal strategies for contracting from perspectives of retailers (i.e., online channel and the offline channel) and manufacturers when retailers offer return service to customers. The corresponding optimal pricing and profit were proposed. Based on the results in Sects. 3 and 4, the customers' sensitivity factor $k$ to pricing and return rate played a significant role in the optimal strategies for retailers and manufacturers in dual channels, which also depended on the cost, the salvage value of returned products, the extra contract fee, and the market segmentation parameter. Without a generality, $C_s > C_o > C > S > 0$ and $0 < W_s < W_o$.

For the offline channel retailer and the manufacturer, the following results are discovered:

If both retailers and manufacturers like the contract, both of them can gain more profits through the contract. Table 1 shows the choices for offline channel retailers and manufacturers regarding contracts and the related conditions. When the retailer provided the offline channel return service to customers, and if both retailer and manufacturer agreed to contract to maximize profits, the optimal solutions of the offline channel are $(P^*_{s2}, \pi^*_{s2})$, otherwise the optimal solutions are $(P^*_{s1}, \pi^*_{s1})$.

Table 2 shows the choices for online channel retailers and manufacturers regarding the contracts to process returned products and the related conditions. Similarly, when retailers provided online channel return service to customers, and when retailers and manufacturers agreed to contact to maximize profits, the optimal solutions for online channel retailers are $(P^*_{02}, \pi^*_{02})$. Otherwise, the optimal solutions are $(P^*_{01}, \pi^*_{01})$.

Tables 1 and 2 show the corresponding conditions, when dual-channel retailers (i.e., a retailer with both an offline and online presence) offered return services to customers or not, and whether the retailer should contract with the manufacturers by paying extra fees to return products to manufacturers. The decision to offer return services depended on expected return probabilities for two channels. This study revealed the beneficial (detrimental) conditions for retailers and manufacturers to contract to handle returned products.

**Table 2** Optimal policies of online channel retailer and manufacturer

| | Retailer likes the contract | Retailer dislikes the contract |
|---|---|---|
| Manufacturer likes the contract | $\begin{cases} 1)\ \text{If } \bar{k}_{01} \le k < \bar{k}_{02} \text{ and } r_{03} < r_0 < \bar{r}_0 \text{ hold} \\ 2)\ \text{If } \bar{k}_{02} \le k < \bar{k}_0 \text{ and } r_{01} < r_0 < \bar{r}_0 \text{ hold} \end{cases}$ | $\begin{cases} 1)\ \text{If } 0 < k < \bar{k}_{02} \text{ and } 0 < r_0 \le r_{01} \text{ hold} \\ 2)\ \text{If } \bar{k}_{01} \le k < \bar{k}_{02} \text{ and } 0 < r_0 \le r_{03} \text{ hold} \end{cases}$ |
| Manufacturer dislikes the contract | $\begin{cases} 1)\ \text{If } 0 < k < \bar{k}_{01} \text{ and } r_{01} \le r_0 \le \bar{r}_0 \text{ hold} \\ 2)\ \text{If } \bar{k}_{01} \le k < \bar{k}_{02} \text{ and } r_{01} < r_0 \le r_{03} \text{ hold} \end{cases}$ | $\begin{cases} 1)\ \text{If } \bar{k}_{02} \le k < \bar{k}_{03} \text{ and } r_{03} < r_0 \le r_{01} \text{ hold} \\ 2)\ \text{If } \bar{k}_{03} \le k < \bar{k}_0 \text{ and } 0 < r_0 \le r_{01} \text{ hold} \end{cases}$ |

**Table 3** Optimal pricing and profit of offline channels under different strategies

| Optimal pricing and profit | Return is not available | Return is available and without a return contract with the manufacturer | Return service is available and contract return with the manufacturer |
|---|---|---|---|
| $P_s^*$ | 0.5833 | $(0.65 - 0.5r_s)/(1 - r_s)$ | 0.7 |
| $\pi_s^*$ | 0.0042 | $(0.15 - 0.3r_s)^2/2(1 - r_s)$ | $0.005(1 - r_s)$ |

**Table 4** Optimal pricing and profit of online channels under different strategies

| Optimal pricing and profit | Return is not available | Return is available and without a return contract with the manufacturer | Return service is available and contract return with the manufacturer |
|---|---|---|---|
| $P_0^*$ | 0.7 | $(0.8 - 0.7r_0)/(1 - r_0)$ | 0.875 |
| $\pi_0^*$ | 0.054 | $(0.4 - 0.5r_0)^2/2(1 - r_0)$ | $0.05281(1 - r_0)$ |

## 6 Case study

This section validates the proposed methods and results via two cases to express the characteristics of the analytic findings and to state related conclusions intuitively and clearly. For simplicity, the total demand is $A = 1$, and $\gamma = 0.4$(resp. $\gamma = 0.6$) is the offline channels segmentation of market demand parameters, and $1 - \gamma = 0.6$(resp. $1 - \gamma = 0.4$) is the online channel segmentation of market demand parameters for two cases. Based on Propositions 2 and 4, the customer's sensitivity factor for pricing played a key role, a value is chosen that the retailer and manufacturer liked for contracting with each other regarding the returned product. The customer's sensitivity factor for pricing, when the return service was not available and when the return service was available, are $b = 0.6$ and $k = 0.5$, respectively. Moreover, $P_S$ and $P_0$ are the pricing for online retailers and offline retailers. Note that $C_s = 0.5$ and $C_0 = 0.4$ are the unit product costs for offline and online retailers, and the manufacturer's cost is $C = 0.3$, the salvage value of the returned product is $S = 0.2$. Note that $W_s = 0.1$ and $W_0 = 0.15$ are the prices raised by manufacturers when offline and online retailers should contract with the manufacturers. Here, $\pi_s$ and $\pi_0$ represent the profits of offline and online retailers; $\pi^m$ represents the profit of manufacture; $r_s$ and $r_0$ represent the return rate of customers in online and offline channels, respectively.

**Case 1: When $\gamma = 0.4$ and $1 - \gamma = 0.6$.**

First, the model in Sect. 3 was used to obtain the following optimal policy and results for offline channels, as shown in Table 3.

Then, the model in Sect. 4 was used to obtain the following optimal policy and results for online channels, as shown in Table 4.

When retailers signed contracts with the manufacturer, the manufacturer charged a higher additional cost for the online channel than for offline channel retailers. The

**Table 5** Optimal policies of offline channel retailer and manufacturer

| Range of return rate | Decisions of offline channel retailers | Manufacturer's Decision |
|---|---|---|
| $0 < r \leq 0.25$ | Do not agree to sign a contract with manufacturers | Do not agree to sign a contract with retailers |
| $0.25 < r \leq 0.5$ | Agree to sign a contract with manufacturers | Agree to sign a contract with retailers |

**Table 6** Optimal policies of online channel retailer and manufacturer

| Range of return rate | Decisions of online channel retailers | Manufacturer's Decision |
|---|---|---|
| $0 < r \leq 0.4286$ | Do not agree to sign a contract with manufacturers | Agree to sign a contract with retailers |
| $0.4286 < r \leq 0.8$ | Agree to sign a contract with manufacturers | Do not agree to sign a contract with retailers |

**Table 7** Optimal pricing and profit of offline channels under different strategies

| Optimal pricing and profit | Return is not available | Return is available and without a return contract with the manufacturer | Return service is available and contract return with the manufacturer |
|---|---|---|---|
| $P_s^*$ | 0.75 | $(0.85 - 0.7r_s)/(1 - r_s)$ | 0.9 |
| $\pi_s^*$ | 0.0375 | $(0.35 - 0.5r_s)^2/2(1 - r_s)$ | $0.045(1 - r_s)$ |

**Table 8** Optimal pricing and profit of online channels under different strategies

| Optimal pricing and profit | Return is not available | Return is available and without a return contract with the manufacturer | Return service is available and contract return with the manufacturer |
|---|---|---|---|
| $P_0^*$ | 0.5333 | $(0.6 - 0.5r_0)/(1 - r_0)$ | 0.675 |
| $\pi_0^*$ | 0.0107 | $(0.2 - 0.3r_0)^2/2(1 - r_0)$ | $0.0078(1 - r_0)$ |

online and offline additional contract costs affected the optimal price and profit, which affects customers' demands or market size. The offline channels segmentation of market demand parameter is $\gamma = 0.4$, and the online channel segmentation of market demand parameter is $1 - \gamma = 0.6$. Comparing the optimal pricing and profit results in Tables 3 and 4, the optimal online pricing is found to be higher than offline, and the profit is also larger. This is because the 1) online marginal cost was lower than offline cost, and the online price was higher than offline price, so the cost advantage and the price advantage made the online channels more profitable; 2) based on our assumptions, more customers choose to buy online.

Following the methods in Proposition 2 and Proposition 4, the optimal decision results for offline channel and online channel are shown in Tables 5 and 6.

Tables 5 and 6 represented retailers' and manufacturers' optimal return policies using various return rates when the retailer provided online and offline return

**Table 9** Optimal policies of offline channel retailer and manufacturer

| Range of return rate | Decisions of offline channel retailers | Manufacturer's Decision |
|---|---|---|
| $0 < r \leq 0.25$ | Do not agree to sign a contract with manufacturers | Agree to sign a contract with retailers |
| $0.25 < r \leq 0.6667$ | Agree to sign a contract with manufacturers | Do not agree to sign a contract with retailers |
| $0.6667 < r \leq 0.7$ | Agree to sign a contract with manufacturers | Agree to sign a contract with retailers |

**Table 10** Optimal policies of online channel retailer and manufacturer

| Range of return rate | Decisions of online channel retailers | Manufacturer's Decision |
|---|---|---|
| $0 < r \leq 0.4286$ | Do not agree to sign a contract with manufacturers | Agree to sign a contract with retailers |
| $0.4286 < r \leq 0.6$ | Agree to sign a contract with manufacturers | Do not agree to sign a contract with retailers |
| $0.6 < r \leq 0.6667$ | Agree to sign a contract with manufacturers | Agree to sign a contract with retailers |

services, as in Case 1. The results are consistent with the research content in Sects. 3 and 4.

**(2) Case 2: When $\gamma = 0.6$ and $1 - \gamma = 0.4$**

Similarly, based on the optimal results in Sect. 3, the following optimal policy and results are obtained for the offline channel, as shown in Table 7.

Then, the optimal results in Sect. 4 were used to obtain the following optimal policy and results for the online channel, as shown in Table 8.

The offline channels segmentation of market demand parameter is $\gamma = 0.6$, and the online channel segmentation of market demand parameter is $1 - \gamma = 0.4$. Comparing the optimal pricing and profit results in Tables 7 and 8, the optimal online pricing is found to be lower than offline, and the profit is also smaller than the offline channel. This is because 1) although the online cost was lower than offline costs, the online price was lower than offline, and the offline price advantage was more remarkable than offline cost disadvantage, so the offline channels were more profitable; 2) more customers choose to purchase in the physical store.

Following the method in Sects. 3 and 4, the optimal decision results for offline channel and online channel retailers are shown in Tables 9 and 10, respectively.

Tables 9 and 10 show the retailers' and manufacturers' return policies using different return rates when online and offline return services are available in Case 2. The results are consistent with the research content in Sects. 3 and 4.

## 7 Conclusions and further work

This paper focused on different customer return probabilities, return losses, and market segmentations among the online and offline channel retailers, on whether the retailers should provide return services to the customer and when provide it, and on whether the retailers should negotiate a contract with the manufacturer and pay additional fees to reduce returned product losses. This work mainly addressed the following aspects: (a) building different profit models for retailers and manufacturers using the online channel and the offline channel perspectives under various conditions; (b) analyzing the related profit functions and obtaining the optimal pricing and optimal profits; (c) analyzing the relationship between the return rate and optimal pricing and selecting optimal strategies in different situations; (d) proposing corresponding optimal solutions and insights.

This study found that dual-channel retailers need to trade-off between stimulus for product demands and the return losses. When the probability of return is small, the dual-channel retailers should provide the return service to the customer to obtain the maximum profit. In this situation, the profit from the product demands increased by offering the return service, and they offset the loss caused by the returns. Then, when the return probability is large, the retailers should cancel the return service offered to customers to reduce the losses caused by undue returns, or they should cancel the sales of products.

It was found that if the dual-channel retailers adopted return services for customers, whether the offline and online retailers and manufacturer contacted each other depended on the customers' sensitivity for pricing and return rates for products. This research showed that when the customers' demands customers were high, the contract desire was only one-sided; however, if customers' demands were low, retailers and manufacturers were likely to reach a consensus. Specifically, if customer's sensitivity for pricing was minor and the return rate was low (resp. high), the manufacturer preferred (resp. dislikes) to contract the manufacturer to recycle the returned products to achieve maximum profit, but the retailer disliked (resp. likes) the contract. If the customer's sensitivity for pricing was large, or if the market size/demand was low, and if the return rate was low, both retailer and manufacturer liked achieving their maximum profit without contract. However, if the return rate was high, the retailer liked the contract to reduce return losses, and the manufacturer earned substantial profit to recycle the returned products for the retailer.

To establish a reasonable and tractable framework and derive insightful results, the assumptions about the demand function and market segmentation were simplified. First, competition between two channels was absent from the model. In the future, how competition may affect the results could be discussed. Second, one channel retailer changed its price, and it will affect the demand of another channel. The interaction between online and offline channels should be examined in future work. In addition, it would be worthwhile to investigate the relationship between customer return rates and prices. Finally, this paper addressed only the interactions between retailers and manufacturers. It would be productive to see how the results would change when the strategies used by retailers and customers are considered.

Therefore, constructing a model based on game-theoretic is another research problem that merits attention.

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